

TreeMMM

Panel attribution within 1–6pp of Oracle baselines, no manual specification required

`pip install treemmm` | MIT License | Working Paper v0.1 | Synthetic benchmarks, single seed

Metric	TreeMMM	GLMM-Naive	GLMM-Oracle	PyMC-Hier-Naive	PyMC-Mktg
Attribution MAPE (non-linear avg)	18.3%	24.0%	17.3%	24.1%	70.7%
Interaction discovery	5 / 6 found	0 / 6	0 / 6 (oracle spec only)	0 / 6 (oracle spec only)	0 / 6
mROI ranking (Spearman rho)	0.96	0.26–1.00	—	—	—
Data granularity	Panel (108K rows)	Panel (108K rows)	Panel (108K rows)	Panel (108K rows)	Aggregate (36 rows)

Customer-Level (Panel) Comparison
TreeMMM within 1–6pp of Oracle; 6pp ahead of Naive baselines

5 of 6 Interactions Found Automatically
(regression finds 0/6 without manual specification)

